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2. PROJECT DESCRIPTION

2.1 Purposes and Project Objectives

2.1.1 The purpose and objective of the Project is to strength the external transport links of the Loop, with constructing a link road between the eastern part of the Loop and Kwu Tung North (KTN) NDA via Ma Tso Lung (MTL) area. Road users can be benefited from this link road travelling between the Loop and other districts in New Territories north such as KTN. It is also identified as an alternative route from / to the Loop to alleviate the traffic impact for Western Connection Road (WCR) of the Loop.

2.2 Location and Scope of the Project

2.2.1 The location and scope of the Project is shown in **60722948/A10/FIGURE 2.1** to **2.5**. The EIA study covers a section within the Loop, a section near Ping Hang, Horn Hill and Tse Koo Hang, a section at the north of Shun Yee San Tsuen at MTL and a section at the east of Ma Tso Lung San Tsuen.

2.2.2 The scope of the Project comprises the construction of connection roads (in form of at-grade road, depressed road, underpass and viaduct) connecting the eastern part of the Loop to KTN NDA via MTL Area, and provision of associated environmental mitigation works, noise mitigation, landscape and other ancillary works. The details are as follows:-

(a) Construction of single 4-lane two-way carriageway, in form of at-grade road, depressed road, underpass and viaduct, of a total length of approximately 2.5km;

(b) Provision of Electrical & Mechanical (E&M) plant room and drainage pumping room to support the operation of depressed road and underpass;

(c) Modification of existing Border Road; and

(d) Associated street furniture, traffic aids, drainage, water, utilities, lightings, electrical and mechanical works, slopes, retaining walls and landscaping works.

2.3 The Need for the Project

2.3.1 The development within the Loop is required to meet the future needs for the development of Hong Kong and Shenzhen as an international innovation and technology hub in the Greater Bay Area. In the Chief Executive's 2007-08 Policy Address, the development of the Loop was identified as one of the ten major infrastructure projects for economic growth and to meet future development needs.

2.3.2 In July 2021, the site formation and infrastructure works, including site formation for the entire Loop, major carriageways / footpath / cycle track within the Loop, was commenced under the Project "Development of the Loop – Main Works Package 1 (MWP1). Located within the boundary area, the WCR, which is constructed under MWP1, currently is the only external transport link to the surrounding areas. When the Phase 2 development of the Loop is implemented, it is anticipated that the WCR will be operated near or above the capacity.

2.3.3 To strengthen the external transport link of the Loop, it is proposed to construct a link road between eastern part of the Loop and the KTN NDA via Ma Tso Lung area. This Project is also considered necessary for supporting Phase 2 development of the Loop

by enhancing transport connectivity and alleviating anticipated traffic congestion. This Project will provide an additional route, distributing traffic more evenly and strengthening the transport network between the Loop and surrounding areas like KTN NDA and MTL.

2.4 Scenarios With and Without the Project

Scenario "without" Project

2.4.1 Without the implementation of this Project, the WCR, which is under construction under "Development of the Loop – Main Works Package 1", would be the only external transport link connecting the Loop and the existing road networks. With implementation of Phase 2 development of the Loop, it is anticipated that the WCR would be operated close to or above capacity. This would lead to long traffic queues to appear along the WCR, consequently leading to higher vehicular emissions from decreased traffic speeds.

Scenario "with" Project

2.4.2 With the Project in place, the traffic demand for the WCR would be shared with the Eastern Connection Road (ECR) and the external transport links between the Loop and the existing road networks would be strengthened by connecting the eastern part of the Loop and the KTN NDA via MTL area. The implementation of the Project results in improvement in the traffic conditions along the WCR. The Project enhances the overall transport system by introducing an alternative route for commuting traffic, thereby relieving congestion on the WCR. Through careful optimisation of the alignment, the Project not only shortens travel time for road users but also contributes to reduced energy consumption and mitigates environmental impacts on nearby communities.

2.5 Consideration of Alignment Options

2.5.1 The alignment of the ECR was initially investigated under the approved Environmental Impact Assessment Report for the Development of Lok Ma Chau Loop (Register Number: AEIAR-176/2013) (hereinafter refer to as "LMCL"). A total of nine alignment options, designated as Options E1 to E9, were evaluated based on key criteria including environmental impact, land resumption requirements, construction cost, and engineering feasibility. These options included of at-grade, viaduct, and tunnel configurations across the meander and fishpond areas near Ping Hang. From an environmental perspective, alignment crossing the meander via the north-eastern side of the Hoo Kok Wai fishpond was not considered due to significant disturbance to adjacent fishponds and result in construction runoff directly affecting nearby water bodies. Alignments that minimize disturbance to fishponds, wetland habitats and the flight line corridor were prioritized. Over these nine option, at-grade, tunnel, and viaduct options across the meander and fishpond near Ping Hang have been reviewed and accessed. Following the comprehensive assessment in the approved EIA Report for the Development of Lok Ma Chau Loop, Option E9 was adopted as the preferred alignment in the approved EIA report.

2.5.2 Option E9 (the shallow underpass scheme) was selected as the preferred alignment because it provides a balanced solution across environmental, engineering, and cost considerations. Environmentally, it minimizes permanent operation impacts on sensitive habitats, particularly the Conservation Area near Hoo Kok Wai, by minimizing direct disturbance to fishponds (e.g., Ponds 36 to 38) and the meander. Unlike tunnel options, E9 maintains a compact design that integrates seamlessly with the Loop's

internal road network, preserving flexibility for public transport arrangements and reducing land-use conflicts. From an engineering perspective, the short underpass length eliminates the need for extensive tunnel infrastructure, reducing construction complexity, energy consumption, and long-term maintenance requirements. Overall, Option E9 achieves an optimal balance between environmental protection, technical feasibility, and project cost, making it the most practical solution.

2.5.3 There are two alignments for the ECR in this EIA, which will be detailed in the following sections. Extensive environmental and engineering considerations have been taken into account during the evaluation of options in order to minimise impact and maximise the benefits as far as possible.

Base Scheme

2.5.4 Base Scheme (i.e. Option E9) is the road scheme studied under Development of Lok Ma Chau Loop EIA Report (Register No.: AEIAR-176/2013). It comprises the construction of depressed road, underpass, viaduct and at-graded carriageway connecting to KTN NDA.

2.5.5 Base Scheme starts to descend and become a depressed road in the front of the ecological area in the Loop. It then changes to underpass underneath and cross the ecological area in the Loop. Then the Base Scheme changes to depressed road, underpass, depressed road and viaduct crossing the stream near Horn Hill. Finally, the Base Scheme changes to at-grade carriageway near Lok Ma Chau Operation Base Lok Ma Chau Division Hong Kong Police Force and connects to KTN NDA via MTL area. The alignment of Base Scheme is shown in **60722948/A10/FIGURE 2.6**.

Option 1 – Depressed Road and Underpass

2.5.6 Option 1 comprises depressed road, underpass, at-graded carriageway and viaduct, which connecting the Loop and the KTN NDA via the MTL area. Option 1 starts to descend and become a depressed road in the front of the ecological area in the Loop. It then changes to underpass underneath and crosses the ecological area in the Loop, existing meander and the ponds. Then the underpass ascends and connects with depressed road near Lok Ma Chau Operation Base Lok Ma Chau Division Hong Kong Police Force. Permanent above-ground structures, such as at-grade roads or viaducts within the ecological area, flight line corridor and fishponds, are avoided to minimize environmental impacts. Unlike the Base Scheme, this option adopts a full underpass across the ecological area, meander, and fishponds rather than a mixed depressed road and underpass configuration. This approach significantly reduces permanent fishpond loss and prevents habitat fragmentation.

2.5.7 Option 1 then connects to the proposed at-grade road near Horn Hill and Tse Koo Hang. After crossing Tse Koo Hang, Option 1 alignment stars to ascends in front of MTL Stream in form of viaduct and cross MTL Stream without modification. At-grade road with slope formation works and viaduct is proposed at the north of Shun Yee San Tsuen. The ECR finally turns south in the form of viaduct and connects to the at-grade road networks of MTL development and KTN NDA. Option 1 also modify the section of existing Border Road near Tse Koo Hang so as to maintain the traffic connection between Border Road and nearby villages. The alignment of Option 1 is shown in **60722948/A10/FIGURE 2.1** to **2.5** and **2.7**.

2.5.8 Consideration had been taken to avoid affecting the compensatory wetlands established under the LMCL project (i.e. Ecological Area (EA) within the Loop and the

Offsite Wetland Compensation Area (OWCA) Area 2) during the road planning and option selection processes. While direct impacts on OWCA Area 2 are fully avoided, the impacts on the LMCL EA are minimised to the maximum practicable extent and any unavoidable impacts resulting from the construction of the ECR will be mitigated during both the construction and operational stages. Any affected compensatory wetlands will be reinstated and established by the project proponent with reference to the Ecological Mitigation/Habitat Creation and Management Plan and the Operation and Maintenance (O&M) Manual submitted under the LMCL project. Detailed reinstatement and monitoring plan will be formulated and submitted to the future management authority before implementation to ensure effectiveness under the project.

Consideration for Each Alignment Options

2.5.9 The environmental and engineering consideration for the alignment options have been compared and presented in **Table 2.1** below.

Table 2.1 Summary of Environmental and Engineering Consideration of Alignment Options

Alignment Options	Considerations	
	Benefits	**Disbenefits**
Base Scheme	• The adoption of an open sunken road and a permanent aboveground corridor in the design helps to minimized disturbance to Eurasian Otter and the bird flight line corridor during operation phase to a certain extent. • Natural ventilation for the underpass to reduce air quality, noise and visual impacts as well as reduce the use of mechanical ventilation. • The alignment has been carefully planned to minimize encroachment into the Conservation Area (CA) Zone, with particular attention given to reducing impacts on the Hoo Hok Wai area.	• Depressed road section near Horn Hill has direct impact to the existing ponds during construction and operation phase. • The at-grade carriageway will affect the natural streams and wetland of the Ma Tso Lung area, the habitat of the Three-banded Box Terrapin during construction phase. • Existing ponds near Ma Tso Lung Blue House will be directly and indirectly affected during construction and operation phase. • Extensive site formation works and road works within the Ma Tso Lung area which are existing wet grassland, watercourses and farmland. • The viaduct may affect the seasonal flight path of night roost during the operation phase. • Have ecological impact during construction phase, more mitigation measures are required. • Cutting is required and result in habitat loss of orchard and shrubland during construction. • During construction, there will be temporary loss of wetland habitats

Alignment Options	Considerations	
	Benefits	**Disbenefits**
		(e.g. ecological area, the meander, fishpond area) and woodland on Horn Hill and Tse Koo Hang, while the road section near Ma Tso Lung will have direct and indirect impact on the stream and wetland. • Potential air and noise impacts on local village such as Shun Yee San Tsuen associated with vehicular traffic. • Presence of junction / run-in along the alignment.
Option 1 – Depressed Road and Underpass Alignment	• The impacts on the sensitive area (e.g., LMCL ecological area) are minimised to the maximum practicable extent and the affected EA would be reinstated, enhanced and maintained upon completion of the road works. • Key permanent impacts on pond habitats and fragmentation impacts at the western end of the alignment now largely avoided through the adoption of an underpass. • Minimized disturbance to Eurasian Otter and the bird flight line corridor during operation phase to the maximum extent possible. • Natural ventilation for the underpass to reduce visual impact as well as reduce the use of mechanical ventilation. • Less land required within the Loop when compared with Base Scheme. • Minimize the loss of existing fishponds near Tse Koo Hang when compare with Base Scheme • Reduce the impact to the woodland / grassland / shrubland at Ma Tso Lung area. • Lower recurrent / maintenance cost compared with Base Scheme. • Shorter construction time compared with Base Scheme. • The alignment has been carefully planned to minimize encroachment	• Have ecological impact during construction phase, mitigation measures are required. • Cutting is required and result in habitat loss of orchard and shrubland. • During construction, there will be a temporary loss of wetland habitats (e.g. ecological area, the meander, fishpond area) and woodland on Horn Hill, while the road section near Ma Tso Lung will have an indirect impact on the wetland. • Potential air and noise impacts on local village such as Shun Yee San Tsuen associated with vehicular traffic. • More natural terrain hazard mitigation works is required. • Extensive site formation works and road works at the north and at the south of Shun Yee Sun Tsuen which are existing wet grassland. • Presence of junction / run-in along the alignment.

Alignment Options	Considerations	
	Benefits	**Disbenefits**
	into the Conservation Area (CA) Zone, with particular attention given to reducing impacts on the Hoo Hok Wai area.	

2.5.10 Having considered the factors as mentioned in **Table 2.1** above and striking the balance between the benefits and disbenefits of each alignment option, Option 1 – Option with Depressed Road and Underpass is concluded to be the most preferred option.

2.5.11 The general layout of the preferred option (i.e. Option 1 – Option with Depressed Road and Underpass) is shown in **60722948/A10/FIGURE 2.1 to 2.5**. Detailed environmental impact assessments of the preferred option are presented in **Sections 3** to **12**.

2.6 Construction Methodologies for Key Design Elements

2.6.1 This Project comprises carriageways (in the form of (i) at-grade section; (ii) depressed road and underpass crossing the ecological area in the Loop, existing meander and existing ponds; and (iii) viaducts near Shun Yee San Tsuen and Ma Tso Lung San Tsuen). The location of depressed road, underpass and viaduct is shown in **60722948/A10/FIGURE 2.1** to **2.5**.

2.6.2 This section describes the planning and sequence of works of the construction of depressed road, underpass and viaduct construction under this Project.

Construction of Depressed Road and Underpass

2.6.3 The ECR starts to descend and become a depressed road in the front of the ecological area in the Loop. In order to minimize the loss of fishponds and reduce the fragmentation effect, it then changes to underpass underneath and cross the ecological area in the Loop, existing meander and existing ponds. Then the underpass ascends and connects with depressed road and at-grade road near Lok Ma Chau Operation Base Lok Ma Chau Division Hong Kong Police Force.

2.6.4 In order to avoid having ventilation buildings for the underpass, two reinforced concrete ventilation openings (about 0.13 ha each) will be provided on the rooftop of the underpass such that natural ventilation can be adopted for the underpass as far as possible to reduce air quality, noise and visual impact and amount of electricity consumed for mechanical ventilation. A ventilation opening including maintenance access (about 0.25 ha) will be created at Pond 38, which can be constructed by backfilling and utilising partial of existing pond bund. The subsequent maintenance of the ventilation opening structures are expected to be managed by the Highway Department.

2.6.5 In exploring construction methods, both cut-and-cover and Tunnel Boring Machine (TBM) were considered to minimize encroachment and direct impacts on the ecological area within the Loop, the existing meander, and adjacent fishponds. TBM was initially reviewed as an alternative to reduce surface disturbance. However, based on available ground investigation records, the alignment of the underpass consists of soft ground without hard rock with high groundwater levels. Implementing TBM would require large launching and retrieval shafts, extensive ground treatment (e.g., grouting or bentonite), and additional land take, which would impose significant impacts on the ecological area

and adjacent natural terrain. From a safety perspective, TBM operations in soft ground with high groundwater levels pose risks of ground settlement and potential flooding during construction, requiring extensive mitigation measures. Given these geotechnical condition, drill-and-break/drill-and-blast method would not be adopted in this Project and thus the use of explosives for construction is not required.

2.6.6 Considering the alignment geometry, geotechnical conditions, and the need to minimize environmental impacts on the existing meander and ponds, the cut-and-cover construction method is considered as a more reliable and controllable approach and recommended to be adopted.

2.6.7 Excavation and Lateral Support (ELS) is an essential component of the cut-and-cover method, as the trench must be stabilized to ensure safe construction of the underpass and depressed roads. This approach manages the stability of excavation walls created during the "cut" phase of cut-and-cover process. In addition, ELS can be created to form a watertight barrier, effectively mitigating groundwater ingress, a particularly valuable feature in areas with high water tables. This dual functionality enhances both safety and construction efficiency. While the detailed construction sequences and processes will be further developed by the contractor, the sequence is illustrated in **Appendix 2.1**.

2.6.8 The cut-and-cover method involves excavating from the ground surface down to the required depth until the base of the tunnel structure is reached. Although there is no direct precedent in adopting the full cut-and-cover method in ecologically sensitive areas, excavation works, which are the major component of cut-and-cover method, have been successfully implemented in similar environments. For instance, excavation works were carried out in ecologically sensitive areas such as the Ecological Area within the Loop, the Shenzhen Meander, and adjacent fishponds under the *Ecological Mitigation/Habitat Creation and Management Plan of Agreement No. CE 5/2014 (CE) – Development of Lok Ma Chau Loop: Land Decontamination and Advance Engineering Works – Design and Construction.* Excavation was undertaken to establish the EA zone, which comprises of reedbeds and freshwater marshes within the Loop. Concurrently, stone columns were installed to enhance ground stability. Following the completion of stone column installation, embankments were constructed using excavated materials sourced from within the Loop. In addition, the construction of the *Western Connection Road* and *Lok Ma Chau Spur Line* involved excavation and piling work, and installation of column for viaduct section adjacent to ecologically sensitive areas such as Shenzhen Meander and fishponds. These examples demonstrate that excavation and lateral support techniques, which form the basis of the cut-and-cover method, can be effectively implemented in ecologically sensitive areas, provided that appropriate mitigation and management measures are in place.

2.6.9 Comparison of the merits and demerits between the construction methods is shown in **Table 2.2** as below.

Table 2.2 Comparison of Different Construction Methods for Depressed Road and Underpass

Construction Methods	Cut-and-Cover	Tunnel Boring Machine (TBM)
Merits	• Conventional construction method and require relatively less special skilled labour force and specialists when	• Less labour required for construction.

Construction Methods	Cut-and-Cover	Tunnel Boring Machine (TBM)
	comparing with other construction method. • Facilitate construction of complex underpass geometry. • Better control complex geotechnical condition. • More flexible for dividing works area within the Site, thus allow flexibility for the contractor to coordinate with other concurrent projects in the vicinity for arrangement of respective construction programme and to minimize cumulative environmental impact. • The impacts on the LMCL EA are minimised to the maximum practicable extent and the affected EA would be reinstated, enhanced and maintained upon completion of the road works.	• TBM is usually non-stop working which can speed up construction programme. • Construction works mainly carried out at underground which can avoid working above ground. • Minimize construction noise and air quality impact during construction by using TBM.
Demerits	• Construction works would be affected by adverse weather conditions. • Temporary occupy of existing meander and ponds would be require, though the affected area will be reinstated.	• Ground treatment by using grouting / bentonite to control adverse effect (adverse effect such as excessive ground settlement, groundwater ingress to the TBM, excessive groundwater drawdown) is required for soft ground tunnelling by using TBM, which would contaminate the existing meander and ponds. • Extra land is required for set up of launching shaft and retrieval shaft. This will impose significant impact to the Loop as the land usage within the Loop is well planned. This will impose significant impact to natural terrain at the other end of the underpass. • Require another trade of skilled workers and additional construction equipment for launching the TBM.

Construction Methods	Cut-and-Cover	Tunnel Boring Machine (TBM)
		• Delivery of TBM would have traffic impact/limitation (such as widening existing road may be required for TBM delivery) to the surrounded road network. • Not suitable for underpass with shallow ground cover. • Not cost effective for short underpass.

Construction of Viaduct

2.6.10 The carriageway section of the ECR near Shun Yee San Tsuen and Ma Tso Lung San Tsuen will be in form of viaduct structure. This is due to the preservation of Shun Yee San Tsuen and Ma Tso Lung San Tsuen and connection to the proposed road networks by concurrent project "Ma Tso Lung Area and Other Sites in Kwu Tung North New Development Area and North District – Zone 1 Ma Tso Lung Area".

2.6.11 The viaduct section near Shun Yee San Tsuen and Ma Tso Lung San Tsuen can be erected by cast in-place concrete construction method, prefabricated steel structure construction method and precast concrete construction method. Having considered the site condition of the proposed viaduct, it is recommended to adopt precast concrete construction method which is a more effective and environmentally friendly approach.

2.6.12 Comparison of the merits and demerits between the construction methods is shown in **Table 2.3** as below.

Table 2.3 Comparison of Different Construction Methods for Viaducts

Construction Methods	Cast In-place concrete	Prefabricated Steel Structure	Precast Concrete
Merits	• Conventional construction method and require relatively less special skilled labour force and specialists when comparing with other construction method. • Durable and less maintenance cost.	• Carry out prefabrication works in fabrication yard away from the site. • Better control of quality and workmanship for works in fabrication yard. • Fabricate the whole bridge deck structure by steel plates / members which does not require formwork, thus less C&D waste / materials will be generated.	• Carry out precasting works in fabrication yard away from the Site. • Better control of quality and workmanship for works in fabrication yard. • Use steel mould as formwork to maximise the reuse of formworks, thus less C&D waste / C&D materials will be generated.

Construction Methods	Cast In-place concrete	Prefabricated Steel Structure	Precast Concrete
		• Minimise concrete mixing truck travelling in site area, thus minimise potential air and noise impact to the surrounding environment. • More efficient construction works as the deck structure can be fabricated off-site concurrently with the substructure works (i.e. allow time overlap of construction activities). • Relatively shorten construction time, bridge deck construction can be non-linear (multiple work fronts), with minimising on-site potential environmental impacts and cumulative environmental impacts with concurrent projects. • Requires relatively less temporary steel platforms, for construction of light weight steel structure.	• Minimise concrete mixing truck travelling in site area, thus minimise potential air and noise impact to the surrounding environmental. • More efficient construction works as the deck segments can be casted off-site concurrently with substructure works (i.e. allow time overlap of construction activities). • Shorten construction time, bridge deck construction can be non-linear (multiple work fronts), with minimising on-site potential environmental impacts and cumulative environmental impacts with concurrent projects. • Durable and less maintenance cost. • Relatively requires less temporary steel platform for construction of precast structure.
Demerits	• Prepare concrete mix at a central concrete batching plant, where control of the materials should be monitored much carefully. • For transporting concrete to the Site, proper mixing in truck, discharging from the truck and depositing	• Subject to high risk of steel corrosion. • High maintenance cost. • Less durable. • Require another trade of skilled workers and additional construction equipment for	• Require another trade of skilled workers and additional construction equipment for erecting precast segments.

Construction Methods	Cast In-place concrete	Prefabricated Steel Structure	Precast Concrete
	in the forms, and handling for placement, finishing and curing are all subject to the level of responsibility and craft exercised by the workers involved. • Site conditions in terms of accessibility and wealth can be highly critical to the work, requirement of extreme measures in some situations to control all stages in the production process. Possibility of site run-off might be higher under bad weather conditions. • Use of timber formwork, thus more C&D waste / C&D materials will be generated. • Longer construction time – carry out bridge deck construction only (1) after substructure works is completed; and (2) after completion of previous span for cast in-place span by span. • Require substantial temporary steel platform / falsework for construction of the bridge deck section over river channel (i.e. Ma Tso Lung channel).	erecting steelwork structures.	

2.6.13 Based on the considerations as listed in **Table 2.3** above, precast concrete construction method is the preferred construction method. The viaduct deck section will be designed to allow for the use of precast concrete construction method. The deck will be formed from precast concrete sections which will be manufactured at a casting yard offsite and

stitched together at their final positions on-site. This approach will minimise the quantities of C&D waste / C&D materials to be generated on-site, and also minimise the extent and duration of construction activities required on-site and hence the potential environmental impacts on nearby sensitive receivers during construction.

2.7 Updates of the Alignment near Ma Tso Lung San Tsuen

2.7.1 Since the issue of EIA Study Brief (No. ESB-375/2025), existing Water Supplies Department (WSD) water tunnel portal and associated washout chamber are identified at the side of Ma Tso Lung Road near Ma Tso Lung San Tsuen. In order to minimise the impact to WSD's facilities and the risk of water supply, this section of ECR is revised from at-grade carriageway to viaduct.

2.8 The Benefits of the Project

2.8.1 This Project will provide a link road to connect the development within the Loop and KTN NDA. The development of the Project could result in the following benefit.

2.8.2 Northern Metropolis (NM) Synergy – According to Northern Metropolis Action Agenda, the Hong Kong-Shenzhen Innovation and Technology Park (the Park) in the Loop is identified as a key area in innovation and technology zone, and KTN NDA is identified as a key area for providing housing supply with more than 80,000 housing units in NM. This Project will act as a key linkage linking up these two major development nodes.

2.8.3 Enhance Traffic Connectivity – Currently, the Western Connection Road is the only external transport link connecting the Loop and the existing road networks. The proposed ECR can share the traffic demand for the Western Connection Road, and can strengthen the connectivity of the Loop and the road networks within KTN NDA via Ma Tso Lung area.

2.8.4 Minimized Environmental Impact – By adopting the preferred alignment of the proposed ECR which incorporates depressed roads and underpasses, the project minimizes disruption to critical habitats including wetlands, fishponds, and corridors for protected species like the Eurasian Otter and migratory birds. The use of natural ventilation for underpasses eliminates the need for energy-intensive mechanical systems, reducing long-term carbon emissions and visual impact. Additionally, the viaduct sections avoid direct impact to Shun Yee San Tsuen. During the operation phase, the ECR alleviates congestion on the WCR, lowering vehicular emissions from idling traffic and shortening travel times and results in further reducing fuel consumption and air quality impact.

2.8.5 Biodiversity Enhancement – Upon completion of the proposed ECR, on-site wetland enhancement would be implemented on the affected ponds (i.e. Ponds 36-38). Wetland associated Avifauna species, non-avifauna species of conservation species as well as Eurasian Otters would benefit from better water management compared to existing pond, and more diverse and shallower sloping edge conditions to provide suitable foraging habitat for these species. Details of the enhancement approaches, including indicative locations and design principles, are presented in **Chapter 8**.

2.9 Project Programme

2.9.1 The construction works of the Project will tentatively commence in Q1/Q2 2027 and completed by Q4 2031. A tentative construction programme for the Project is shown in **Appendix 2.2**.

2.9.2 During construction, the ECR will be divided into three main sections: the first section extends from the depressed road at the Loop to the end of the underpass near the Lok Ma Chau Operational Base (Section 1); the second section covers the depressed road near Lok Ma Chau Operational Base, transitioning to an at-grade alignment along Border Road before converting into a viaduct near Shun Yee San Tsuen (Section 2); and the third section comprises the remaining at-grade alignment at Shun Yee Shan Tsuen, continuing to the viaduct near Ma Tso Lung San Tsuen (Section 3). For planning purposes, construction is assumed to proceed in two phases: Phase 1 will include Sections 1 and 3, while Phase 2 will include Sections 2 and 3. In general, Sections 1 and 2 will not be constructed concurrently, whereas Section 3 may be undertaken throughout the entire construction period. The indicative layout is shown in **60722948/A10/FIGURE 2.8**.

2.10 Concurrent Projects

2.10.1 Concurrent projects in the vicinity of the Site are identified. The status of these concurrent projects is based on the available information obtained during the course of the EIA Study. It should be noted that the implementation of individual project would be subject to further development and subsequent actions of the respective project proponent.

2.10.2 The “Development of the Loop”, which is a designated project, comprises the developments and infrastructure within the Loop and the adjacent areas in Hong Kong. The Loop will be developed for higher education as the leading land use with some elements of new high-tech research and development facilities and creative industries. The construction works is commenced in 2021 for expected completion in phases in around 6 years. While for the buildings within the Loop under construction by the Park, the construction works is commenced in 2024 and will be fully completed by 2035.

2.10.3 The “Ma Tso Lung Area and Other Sites in Kwu Tung North New Development Area and North District – Zone 1 Ma Tso Lung Area”, which is a designated project, is to development the sites at Ma Tso Lung area for residential and employment uses, supporting G/IC facilities, open space and/or amenities. Based on the available information at the time of this EIA study, the construction works of the project is tentatively scheduled to commence in 2028 and the occupation / population intake in 2034.

2.10.4 The “Northern Metropolis Highway – Kwu Tung Section”, which is a designated project, is to enhance the connectivity between the interchange at Fanling and the proposed interchange at Man Kam To Road to alleviate the future traffic demands generated by the future developments in the concerned locations. Based on the available information at the time of this EIA study, the project is expected to commission in about 3 to 4 years after the commissioning of the San Tin Section, in or before 2036, to alleviate the potential traffic congestion at Fanling Highway as well as to support New Development Areas.

2.10.5 The proposed Development of Wetland Conservation Parks under the Northern Metropolis Development Strategy covers a wide area of wetland habitats in the Northwest New Territories area. Within the proposed Hoo Hok Wai (HHW) Wetland Conservation Park (WCP), the existing network of fishponds will largely be retained but zoned and managed to enhance the area’s ecological capacity while promoting the sustainable modernisation of aquaculture practices. Rural activities such as leisure farming may also be introduced to allow visitors to experience a rural lifestyle and enjoy the idyllic countryside landscape. The proposed HHW WCP overlaps with the current

Project Assessment Area, to the north of the Project Alignment. The establishment of the WCP is anticipated to improve the overall quality of the wetland habitats WCP System Area, including the wetland habitats at HHW WCP. The Project would be expected to establishment prior to the detailed studies for the HHW WCP. Potential interface issues will be further reviewed and addressed at the detailed studies stage as indicated in *Strategic Feasibility Study on the Development of Wetland Conservation Parks System Under the Northern Metropolis Development Strategy*. While no detailed information about the proposed Development of Wetland Conservation Parks was available at the time of preparing this EIA Report, potential cumulative ecological impacts have been considered in **Sections 8** of this EIA Report based on the best available information.

2.10.6 Potential cumulative environmental impact arising from these concurrent projects during construction and operation phases of the Project have been assessed in the EIA Report. For construction phase, due to temporal overlapping of construction works of the Project and the "Development of the Loop" and "Ma Tso Lung Area and Other Sites in Kwu Tung North New Development Area and North District – Zone 1 Ma Tso Lung Area", cumulative environmental impact would be anticipated. The potential cumulative impacts are presented in **Sections 3, 4, 5, 8** and **10**. No cumulative impact is expected for other environmental aspects, such as waste management implication, land contamination and cultural heritage. The location plan of concurrent projects is presented in **60722948/A10/FIGURE 2.9**.

2.11 Documentation of Public Concerns

2.11.1 During the public inspection of the Project Profile (No. PP-685/2025) held from 15 May 2025 to 28 May 2025 and 10 June 2025 to 23 June 2025, several public comments were received.

2.11.2 Consultation with the North District Council and Yuen Long District Council was carried out at its meeting held on 8 July 2025 and 29 July 2025, respectively. The Project was well received by the members of both District Councils.

2.11.3 The Project design has taken into consideration the public views and concerns, and mitigation measures have been recommended as appropriate to minimise potential environmental impacts. The key views / concerns and the corresponding design considerations and follow up actions taken are summarised in **Table 2.4**.

Table 2.4 Summary of Key Public Concerns and Subsequent Actions Taken in Response

Item No.	Key Concerns	Follow-up Actions to be Taken
1.	Potential ecologically impacts on ecologically sensitive sites and avifauna	The selected alignment has been optimized to avoid temporary and permanent impacts on the ecological sensitive areas (e.g. LMCL EA) as well as fragmentation impacts at the western end of the alignment with adoption of an underpass at maximum practicable extent. Construction works would be phased in the ecologically sensitive area to ensure some areas of intact habitat are maintained at all times. Enhancement of affected fishpond, reinstatement of the LMCL EA to compensate for ecological functions loss arising from the Project. The LMCL EA shall be enhanced and maintained upon completion to no less than its existing condition. During construction and operation phases, mortality and disturbance to avifauna will be reduced through proposed mitigation measures including the use of quality powered equipment, noise barrier, restricted working hours of aboveground construction in wetland areas, and minimisation of night-time lighting. Details are available of Chapter 8 of this EIA report.
2.	Potential ecologically impacts on wildlife movement in particular the Eurasian Otter	To minimise the potential fragmentation impact, construction works in the ecological sensitive sites will be phased so that some intact habitat is maintained are maintained at all times. In addition, a temporary aboveground wildlife crossing will be provided in the fishpond near Ngau Kok Shan to facilitate the movement of non-flying mammals of conservation importance such as Eurasian Otter. The fishponds in this area would be reinstated and enhanced, thereby avoiding long-term fragmentation effects. Along other sections of the alignment where the ECR follows the

Item No.	Key Concerns	Follow-up Actions to be Taken
		existing Border Road, fragmentation during construction is considered relatively minor as works will not be undertaken simultaneously along the entire route. In the operation phase, fragmentation impacts will be further reduced through specific design features including wildlife underpass with solid parapet. Details are available of Chapter 8 of this EIA report.
3.	Potential ecologically impacts on light pollution	During construction, potential lighting impacts will be minimised by eliminating unnecessary illumination and orienting lights downward into works areas to reduce glare and light spill into adjacent habitats. For long-term operation, road lighting along both at-grade and viaduct sections of the ECR will be carefully planned and designed to minimise disturbance to surrounding habitats including limiting the number of streetlights to the minimum required for safety, reducing lighting intensity to the lowest acceptable level, adopting light sources with narrower spectral wavelengths rather than broad-spectrum white light, and shielding fixtures to direct light onto the roadway while avoiding trespass into sensitive wetland habitats. Details are available of Chapter 8 of this EIA report.
4.	Potential ecological impact and mitigation measures to Fishpond No. 36 to 38	The selected alignment has been optimized to avoid permanent impacts on the ecological sensitive areas as well as fragmentation impacts at the western end of the alignment with adoption of an underpass as far as possible. Ponds 36 to 38 will be reinstated and ecologically enhanced for long-term ecological use upon completion of the ECR. This enhancement represents an additional ecological merit, as it will provide permanently improved habitat quality and

Item No.	Key Concerns	Follow-up Actions to be Taken
		functionality within the Deep Bay Wetland System. Details are available of Chapter 8 of this EIA report.
5.	Potential noise impact and mitigation measures to existing nearby villages	For the construction noise, with the implementation of the proposed mitigation measures, construction noise impact to the Noise Sensitive Receivers (NSRs) could be effectively controlled. Based on the assessment findings for the road traffic noise impact, with the implementation of direct noise mitigation measures of Low Noise Road Surfacing (LNRS) and at-receiver mitigation measures for planned NSRs, such as acoustic window/balconies, the predicted road traffic noise levels at all noise sensitive receivers would comply with noise criteria. Regarding the potential fixed noise source, the design of pumping station is fully enclosed with the exhaust louver oriented to the north, facing Hoo Hok Wai. This configuration is intended to minimize the impact of potential fixed noise sources. No residual impact would be expected from fixed noise source with proper implementation of the recommended mitigation measures. Details are available of Chapter 4 of this EIA report.

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